

TABLE II
Memory Forensics Tools

Name	Platform	Description
CMAT	Windows	Memory-analysis tool that analyses Windows OS memory and extracts information about the OS and the running processes (www.sourceforge.net/projects/cmat)
Memoryze	Windows	Memory forensic software that helps incident responders find evi in live memory (www.mandiant.com/products/free_software/memoryze)
Responder	Windows	(www.hbgary.com/responder-field) and (www.hbgary.com/responder-pro)
Second Look	Linux	Easy-to-use memory acquisition and analysis capabilities for Linux system (www.secondlookforensics.com)
WindowsSCOPE	Windows	WindowsSCOPE is the next generation in live cyber forensics tools and memory forensics technologies (www.windowsscope.com)
Volafox	Mac OS	Volafox or Mac OS X Memory Analysis Toolkit is developed on Python 2.x (www.code.google.com/p/volafox)
Volatility	Window, Linux	A completely open collection of tools, implemented in Python under GNU general-public licence, for extraction of digital artefacts from volatile memory (RAM) samples (www.code.google.com/p/volatility)

TABLE III
Hex Editors

Tool name	Platform	Description
Okteta	Linux	A simple editor for raw data of files. This type of program is also called hex editor or binary editor (www.utils.kde.org/projects/okteta)
Hex Dump	Linux, Windows	A hexadecimal view (on screen or paper) of computer data, from RAM or from a file or storage device (www.linux.about.com/library/cmd/bicmd11_hexdump.htm)
Hex Fiend	Mac OS X	A hex editor for Apple OS X (Fig. 4) (www.nidiculousfish.com/hexfiend)
Hex Workshop	Windows	Allows you to easily view, dissect and modify binary data (www.bpssoft.com)
KHEdit	Linux	A versatile binary file editor; key features include undo/redo, overwrite/insert modes, hexadecimal, octal, binary or text only display, etc (www.freecode.com/projects/khexedit)
WinHex	Windows	A universal hexadecimal editor, particularly helpful in the realm of computer forensics, data recovery, low-level data processing and IT security. An advanced tool for everyday and emergency use; inspects and edits all kinds of files, recovers deleted files or lost data from hard drives with corrupt file systems or from digital camera cards (Fig. 5) (www.winhex.com/winhex)
xxd	Linux	A command utility that creates a hex dump of a given file or standard input; can also convert a hex dump back to its original binary form (www.linuxcommand.org/man_pages/xxd.1.html)
HxD	Windows	HxD Hex Editor provides tools to inspect and edit files, main memory, disks/disk images and their structure using a simple and modern interface (download.cnet.com/HxD-Hex-Editor/3000-2352_4-10891068.html)

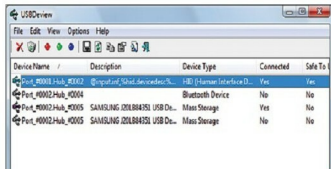


Fig. 3: USBDeview details previously attached USB devices

Computer memory forensics

Memory forensic tools are used to acquire and/or analyse a computer's volatile memory, that is, RAM. These are often used in incident response situations to preserve evidence in memory that would be lost if a system is shut down and to quickly detect stealthy malware by directly examining the OS and others

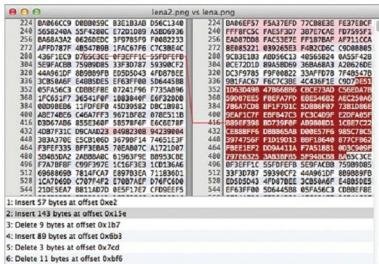


Fig. 4: Hex Fiend, a hex editor for Mac OS X

running the software in memory. The various computer memory forensics tools are given in Table II.